

j	14	15	16	22	24	25
$N(j)$	1	1	3	2	2	5

$$N = \sum_j N(j) = 14$$

a) $\langle j^2 \rangle = ?$, $\langle j \rangle^2 = ?$

$$\begin{aligned} \langle j^2 \rangle &= \frac{\sum_j j^2 N(j)}{N} = \frac{1}{14} \left(14^2 + 15^2 + 3 \cdot 16^2 + 2 \cdot 22^2 + 2 \cdot 24^2 + 5 \cdot 25^2 \right) \\ &= \frac{1}{14} \left(196 + 225 + 768 + 968 + 1152 + 3125 \right) \\ &= \dots \text{calculator} \dots = \frac{1}{14} \cdot 6434 \end{aligned}$$

$$\begin{aligned} \langle j \rangle &= \frac{\sum_j j N(j)}{N} = \frac{1}{14} \left(14 + 15 + 3 \cdot 16 + 2 \cdot 22 + 2 \cdot 24 + 5 \cdot 25 \right) \\ &= \frac{1}{14} \left(14 + 15 + 48 + 44 + 48 + 125 \right) \\ &= \frac{294}{14} = 21 \end{aligned}$$

$$\Rightarrow \langle j \rangle^2 = 21^2$$

b) $\sigma^2 = \frac{\sum_j (j - \overset{21}{\uparrow} \langle j \rangle)^2 N(j)}{N} = \frac{1}{14} \left((14-21)^2 + (15-21)^2 + 3 \cdot (16-21)^2 \right. \\ \left. + 2 \cdot (22-21)^2 + 2 \cdot (24-21)^2 + 5 \cdot (25-21)^2 \right)$

$$\begin{aligned} &= \frac{1}{14} \left(7^2 + 6^2 + 3 \cdot 5^2 + 2 \cdot 1^2 + 2 \cdot 3^2 + 5 \cdot 4^2 \right) \\ &= \frac{1}{14} \left(\frac{49+36+75+2+18+80}{140} \right) \\ &= \frac{260}{14} \Rightarrow \sigma = \sqrt{\frac{260}{14}} \end{aligned}$$

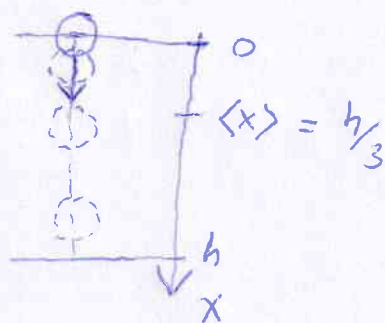
c) Verifying whether $\langle p^2 \rangle - \langle p \rangle^2$ results in the same σ^2 :

$$\begin{aligned}\sigma^2 &= \langle p^2 \rangle - \langle p \rangle^2 = \frac{6434}{14} - \frac{294 \cdot 21}{14} \\&= \frac{6434 - (6300 - 126)}{14} \\&= \frac{134 + 126}{14} = \frac{260}{14} \rightarrow \text{same as (b)}\end{aligned}$$

1.21

Falling rock: $\begin{cases} 0 \leq x \leq h \\ x = \frac{1}{2} g t^2 \end{cases}$

$$P(x) = \frac{1}{2\sqrt{hx}}$$



a) $\sigma = ?$

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\langle x^2 \rangle - \frac{h^2}{9}}$$

$$\hookrightarrow \langle x^2 \rangle = \int_0^h x^2 P(x) dx = \frac{1}{2\sqrt{h}} \int_0^h x^{3/2} dx$$

$$= \frac{1}{2\sqrt{h}} \left[\frac{x^{5/2}}{5/2} \right]_0^h = \frac{1}{2\sqrt{h}} \left(\frac{2}{5} h^{5/2} - 0 \right) = \frac{h^2}{5}$$

$$\Rightarrow \sigma = \sqrt{\frac{h^2}{5} - \frac{h^2}{9}} = h \sqrt{\frac{9-5}{45}} = \frac{2h}{3\sqrt{5}}$$

b) Probability to be outside interval $[\langle x \rangle - \sigma, \langle x \rangle + \sigma]$: P_{outside}

$$P_{\text{outside}} = 1 - P_{\text{inside}}$$

$$\hookrightarrow P_{\text{inside}} = \int_{\langle x \rangle - \sigma}^{\langle x \rangle + \sigma} P(x) dx$$

$$= \frac{1}{2\sqrt{h}} \int_{\langle x \rangle - \sigma}^{\langle x \rangle + \sigma} x^{-1/2} dx$$

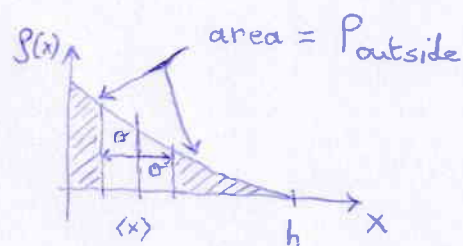
$$= \frac{1}{2\sqrt{h}} \left[\sqrt{x} \cdot 2 \right] \bigg|_{\frac{h}{3} - \frac{2h}{3\sqrt{5}}}^{\frac{h}{3} + \frac{2h}{3\sqrt{5}}}$$

$$= \sqrt{\frac{1}{3} + \frac{2}{3\sqrt{5}}} - \sqrt{\frac{1}{3} - \frac{2}{3\sqrt{5}}}$$

$$= \dots \text{calculator} \dots \approx 0.607$$

$$P_{\text{outside}} \approx 0.393$$

$$\begin{matrix} \downarrow & \downarrow \\ h/3 & \frac{2h}{3\sqrt{5}} \end{matrix}$$



1.3/

$$P(x) = A e^{-\lambda(x-a)^2}$$

 $A, \lambda, a > 0$ and real ($\in \mathbb{R}$)

Formulas integrals:

exponential: $\int_0^{+\infty} x^n e^{-x/a} dx = n! a^{n+1}$

Gaussian:
$$\begin{cases} \int_0^{+\infty} x^{2n} e^{-x^2/a^2} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{a}{2}\right)^{2n+1} \\ \int_0^{+\infty} x^{2n+1} e^{-x^2/a^2} dx = \frac{n!}{2} a^{2n+2} \end{cases}$$

a) Normalization constant $A = ?$

$$1.16 \rightarrow \int_{-\infty}^{+\infty} P(x) dx \stackrel{!}{=} 1$$

$$1 = \int_{-\infty}^{+\infty} A e^{-\lambda(x-a)^2} dx$$

$$= A \int_{-\infty}^{+\infty} e^{-\lambda u^2} du$$

Substitution
 $\begin{cases} u = x - a \\ du = dx \end{cases}$

$$= A \cdot 2 \cdot \int_0^{+\infty} e^{-u^2/(\frac{1}{\lambda})} du$$

$$= A \cdot 2 \cdot \sqrt{\pi} \frac{(2 \cdot 0)!}{\cancel{0!}} \left(\frac{\frac{1}{\sqrt{\lambda}}}{2} \right)^{2 \cdot 0 + 1}$$

$$= A \frac{\sqrt{\pi}}{\sqrt{\lambda}} \Rightarrow A = \sqrt{\frac{\lambda}{\pi}}$$

b) $\langle x \rangle$, $\langle x^2 \rangle$, and $\sigma = ?$

$$\langle x \rangle = \int_{-\infty}^{+\infty} x P(x) dx = A \int_{-\infty}^{+\infty} x e^{-\lambda(x-a)^2} dx$$

$$= A \int_{-\infty}^{+\infty} (u+a) e^{-\lambda u^2} du = \overset{\text{symmetry}}{\downarrow} 0 + A \cdot a \cdot \int_{-\infty}^{+\infty} e^{-\lambda u^2} du$$

$$= A \cdot a \cdot 2 \cdot \frac{1}{2} \sqrt{\frac{\pi}{\lambda}} = \sqrt{\frac{\lambda}{\pi}} \cdot a \cdot \sqrt{\frac{\pi}{\lambda}} = a$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 \rho(x) dx = A \int_{-\infty}^{+\infty} x^2 e^{-\lambda(x-a)^2} dx$$

$$= A \int_{-\infty}^{+\infty} (u+a)^2 e^{-\lambda u^2} du \quad \begin{cases} u = x-a \\ du = dx \end{cases}$$

$$= A \int_{-\infty}^{+\infty} (u^2 + 2au + a^2) e^{-\lambda u^2} du$$

$$= A \int_{-\infty}^{+\infty} u^2 e^{-\lambda u^2} du + 2a A \underbrace{\int_{-\infty}^{+\infty} u e^{-\lambda u^2} du}_0 + a^2 A \underbrace{\int_{-\infty}^{+\infty} e^{-\lambda u^2} du}_1$$

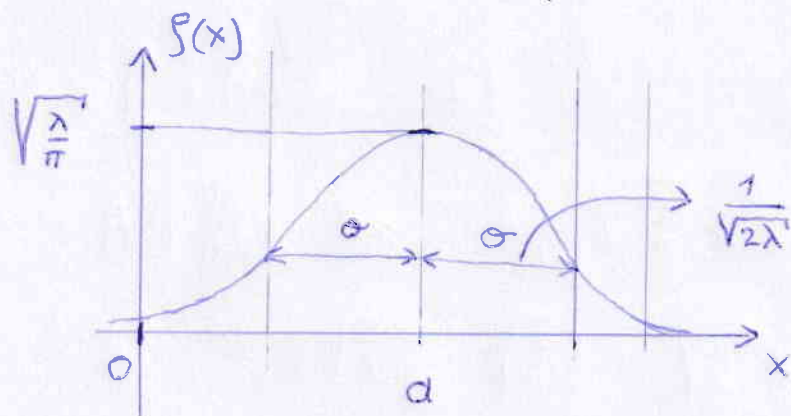
$$= A \int_{-\infty}^{+\infty} u^2 e^{-\lambda u^2} du + \cancel{2aA \cdot 0} + a^2 \cdot 1$$

$$= A \frac{2\sqrt{\pi}}{1!} \frac{2!}{(2\sqrt{\lambda})^3} + a^2$$

$$= A \frac{\sqrt{\pi}}{2\lambda^{3/2}} + a^2 = \sqrt{\frac{\lambda}{\pi}} \frac{\sqrt{\pi}}{2\sqrt{\lambda}\lambda} + a^2 = \frac{1}{2\lambda} + a^2$$

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda} + a^2 - a^2 = \frac{1}{2\lambda}$$

c)



1.5

$$\Psi(x, t) = A e^{-\lambda|x|} e^{-i\omega t}$$

 $A, \lambda, \omega > 0$ and real

a) Normalization constant $A = ?$

$$\begin{aligned} 1 &= \int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = \int_{-\infty}^{+\infty} |\Psi(x)|^2 dx \\ &= |A|^2 \int_{-\infty}^{+\infty} e^{-2\lambda|x|} dx = A^2 \cdot 2 \cdot \int_0^{+\infty} e^{-2\lambda x} dx \\ &= A^2 \cdot 2 \cdot \cancel{0!} \left(\frac{1}{2\lambda} \right)^1 = A^2 \frac{2}{2\lambda} \end{aligned}$$

$$\Rightarrow A = \sqrt{\lambda}$$

b) $\langle x \rangle, \langle x^2 \rangle = ?$

$$\langle x \rangle = A^2 \int_{-\infty}^{+\infty} x e^{-2\lambda|x|} dx = A^2 \cdot \overset{\text{Symmetry}}{\downarrow} 0 = 0$$

$$\begin{aligned} \langle x^2 \rangle &= A^2 \int_{-\infty}^{+\infty} x^2 e^{-2\lambda|x|} dx = A^2 \cdot 2 \cdot \int_0^{+\infty} x^2 e^{-2\lambda x} dx \\ &= \lambda \cdot 2 \cdot 2 \cdot \frac{1}{(2\lambda)^3} = \frac{1}{2\lambda^2} \end{aligned}$$

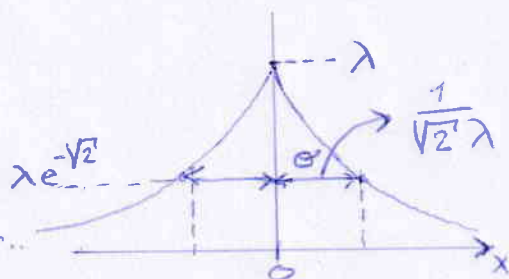
c) $\sigma = ?$, sketch $P(x)$, and find P_{outside} , the probability to be outside interval $[\langle x \rangle - \sigma, \langle x \rangle + \sigma]$

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda^2} - 0 = \frac{1}{2\lambda^2}$$

$$\Rightarrow \sigma = \frac{1}{\sqrt{2}\lambda}$$

$$P(x) = |\Psi|^2 = |\sqrt{\lambda} e^{-\lambda|x|} e^{-i\omega t}|^2 = \lambda e^{-2\lambda|x|}$$

$$P(\pm\sigma) = \lambda e^{-(2\lambda \frac{1}{\sqrt{2}\lambda})} = \lambda e^{-\sqrt{2}} \approx \dots \text{calculator}$$



$$\begin{aligned} P_{\text{outside}} &= 1 - P_{\text{inside}} = 1 - 2\lambda \int_0^{\sigma} e^{-2\lambda x} dx = 1 - 2\lambda \left[\frac{e^{-2\lambda x}}{-2\lambda} \right] \Big|_0^{\sigma} \\ &= 1 + (e^{-2\lambda\sigma} - e^{-2\lambda\sigma}) = e^{-2\lambda\sigma} = e^{-\sqrt{2}} \approx \dots \text{calc.} \end{aligned}$$

1.8.1

$$\begin{cases} i\hbar \frac{\partial}{\partial t} \psi = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V \psi \\ i\hbar \frac{\partial}{\partial t} \psi_0 = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi_0 + (V + V_0) \psi_0 \end{cases} \quad \text{with } \psi_0 = \underbrace{\psi e^{-iV_0 t/\hbar}}_{\text{assumption}}$$

Verify assumption by filling it in in the TDSE

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \left(\underbrace{\psi}_{\psi_0} \cdot e^{-iV_0 t/\hbar} \right) &= i\hbar \left(\frac{\partial \psi}{\partial t} \right) \cdot e^{-iV_0 t/\hbar} + i\hbar \psi \cdot \left(-\frac{iV_0}{\hbar} \right) \cdot e^{-iV_0 t/\hbar} \\ &= i\hbar \left(\frac{\partial \psi}{\partial t} \right) \cdot e^{-iV_0 t/\hbar} + V_0 \psi e^{-iV_0 t/\hbar} \\ &= \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi \right) \cdot e^{-iV_0 t/\hbar} + V_0 \psi \cdot e^{-iV_0 t/\hbar} \\ &= -\frac{\hbar^2}{2m} \frac{\partial^2 \psi_0}{\partial x^2} + V \psi_0 + V_0 \psi_0 \end{aligned}$$

ψ_0 is a solution of the Schrödinger equation 